

What is LAMMPS, and what is this course?

LAMMPS is a classical molecular dynamics engine maintained by Sandia/NTESS: you write an `input script`, it integrates Newton's equations for a box of particles under a chosen potential.

- Two release lines track upstream: *develop* (daily) and one **stable** release a year with periodic updates.
- This course sits on top of `lammps-kill` (a verified toolkit: script builder, checker, parsers, runner) and `mdlite` (a small teaching engine that reproduces LAMMPS's numbers from scratch).
- Every chapter follows **record-or-run**: with LAMMPS installed, the computation runs live; without it, a stored result loads instead -- the figures never lie about which happened.

First Simulation

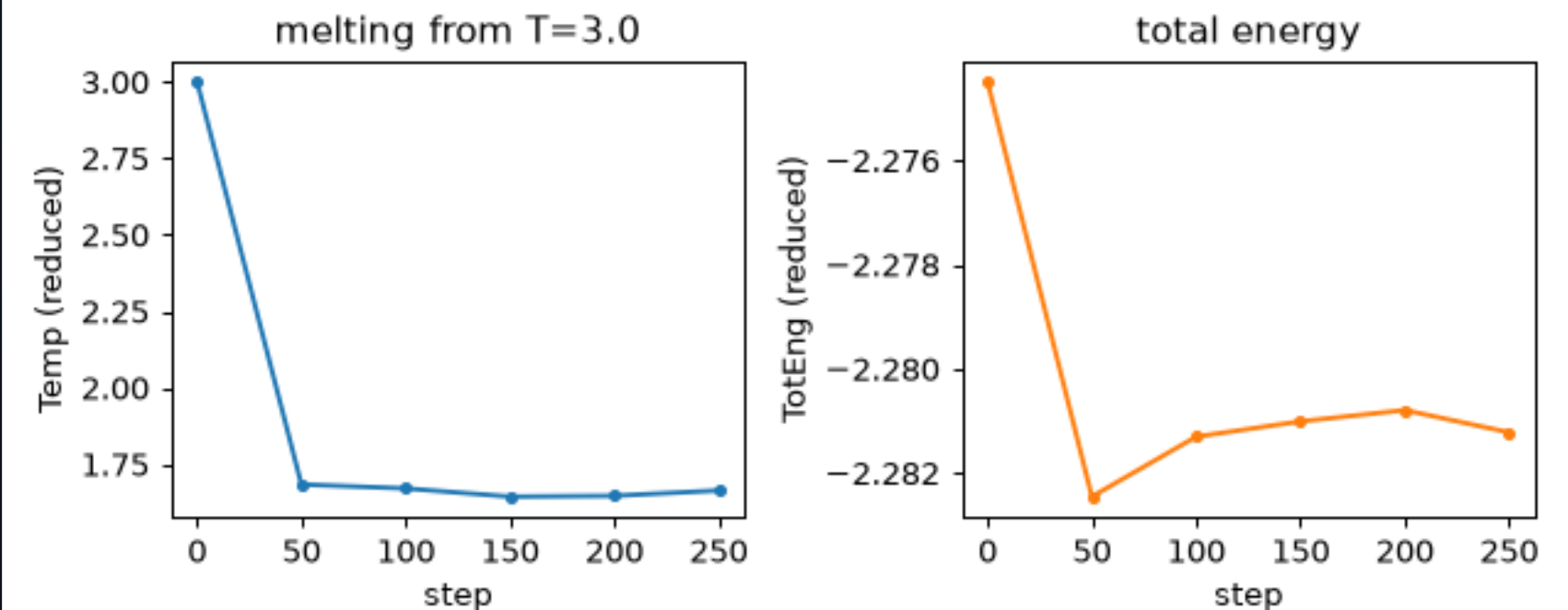
A Lennard-Jones melt end to end -- Spec, check, run, read the log, plot -- through the checker's fourteen rules, one at a time.

1 One simulation, start to finish

One simulation, start to finish

`Spec` describes the case in Python; `check` applies fourteen rules against the manual before anything runs; `run` calls LAMMPS; the log comes back as a typed `ThermoRun`.

- The case: an fcc Lennard-Jones lattice heated until it melts, in reduced (LJ) units.
- Nothing here is a hand-typed input script -- the checker exists so a beginner's script fails before LAMMPS ever sees it.
- The same four steps -- build, check, run, read -- reappear in every later chapter.



The LJ melt's thermo history read back from the log: temperature falling away from the $T=3.0$ start (left) while the total energy holds flat (right) -- the two curves `run()` reads out of LAMMPS's own thermo output, whether this cell just ran the case or loaded a stored record. — Figure 1 · notebook §0 · chapter 1, cell 12

Forces and Integration

Velocity Verlet in mdlite, forces checked against a numerical derivative, and energy conservation against the timestep, cross-checked with LAMMPS at $3\text{e-}13$.

1 Integration is the part that must not be approximate

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`mdlite` implements velocity Verlet from the same equations LAMMPS integrates, so its forces can be checked against a numerical derivative and against LAMMPS itself.

- Forces from `mdlite`'s Lennard-Jones match a central-difference numerical derivative to machine precision.
- Against a real LAMMPS run of the identical case, the two energies agree to 3×10^{-13} -- there is no approximation hiding in either implementation.
- Energy conservation versus timestep is the practical version of the same idea: too large a step and the trajectory drifts, long before it visibly explodes.

Thermostats

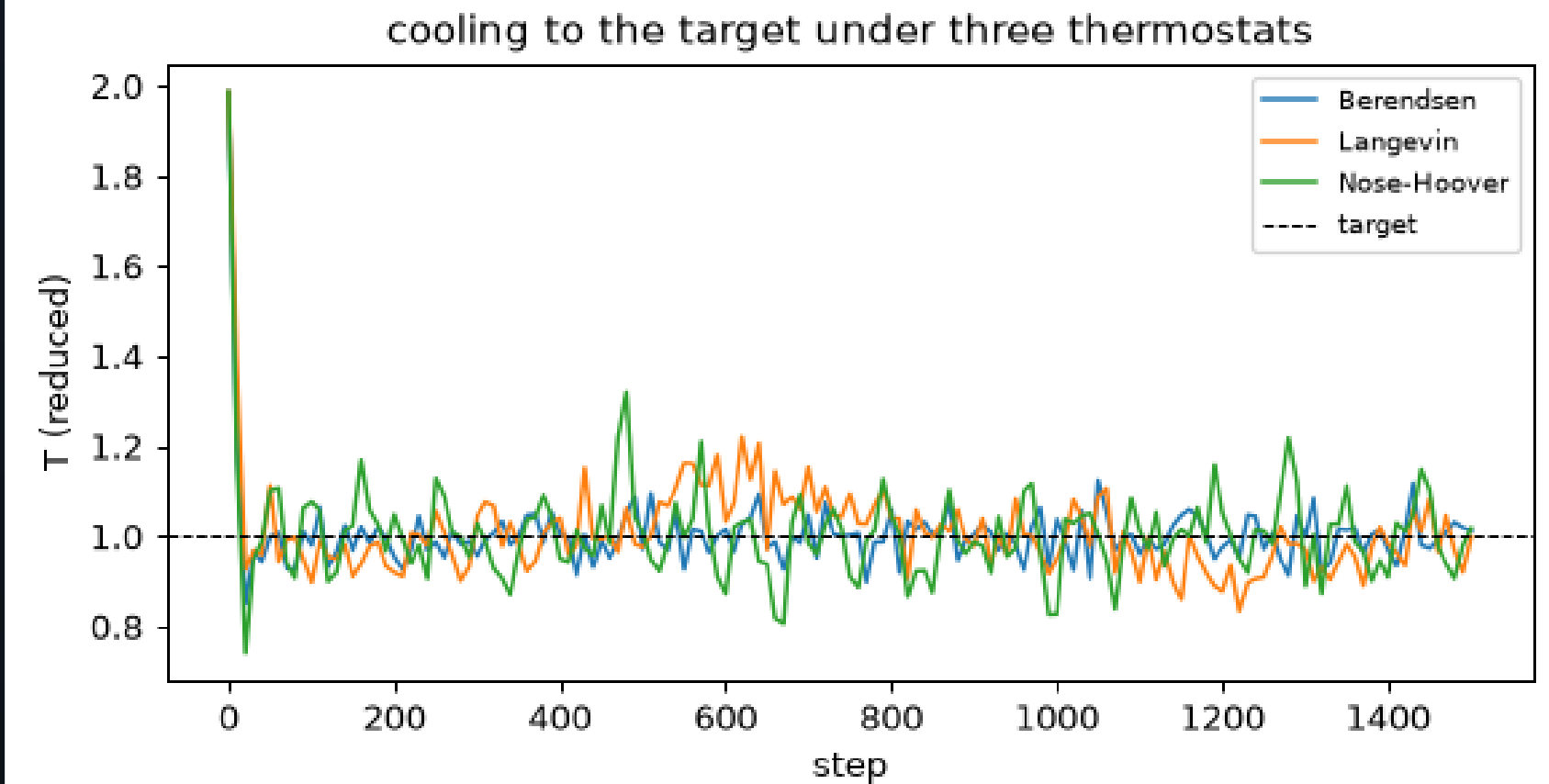
Berendsen, Langevin and a Nose-Hoover chain: what each one conserves, what each distorts, and NVT checked against the NIST reference within 3 sigma.

1 Three ways to hold a temperature

Three ways to hold a temperature

A thermostat couples the system to an external bath; which one you choose changes what else it disturbs.

- Berendsen: fast, simple, but does not sample the correct canonical ensemble -- fine for equilibration, not for production statistics.
- Langevin: adds friction and noise per atom; correct ensemble, but its own set of fluctuations to tune.
- Nose-Hoover chain: an extra dynamical variable keeps the total (system + thermostat) energy conserved -- checked here directly.



Temperature against step for the same fcc lattice, started hot ($T \sim 2$) and cooled by three thermostats -- Berendsen, Langevin, Nose-Hoover chain -- each converging toward the target (dashed line) by its own route. — Figure 1 · notebook §0 · chapter 3, cell 4

Ensembles and Restarts

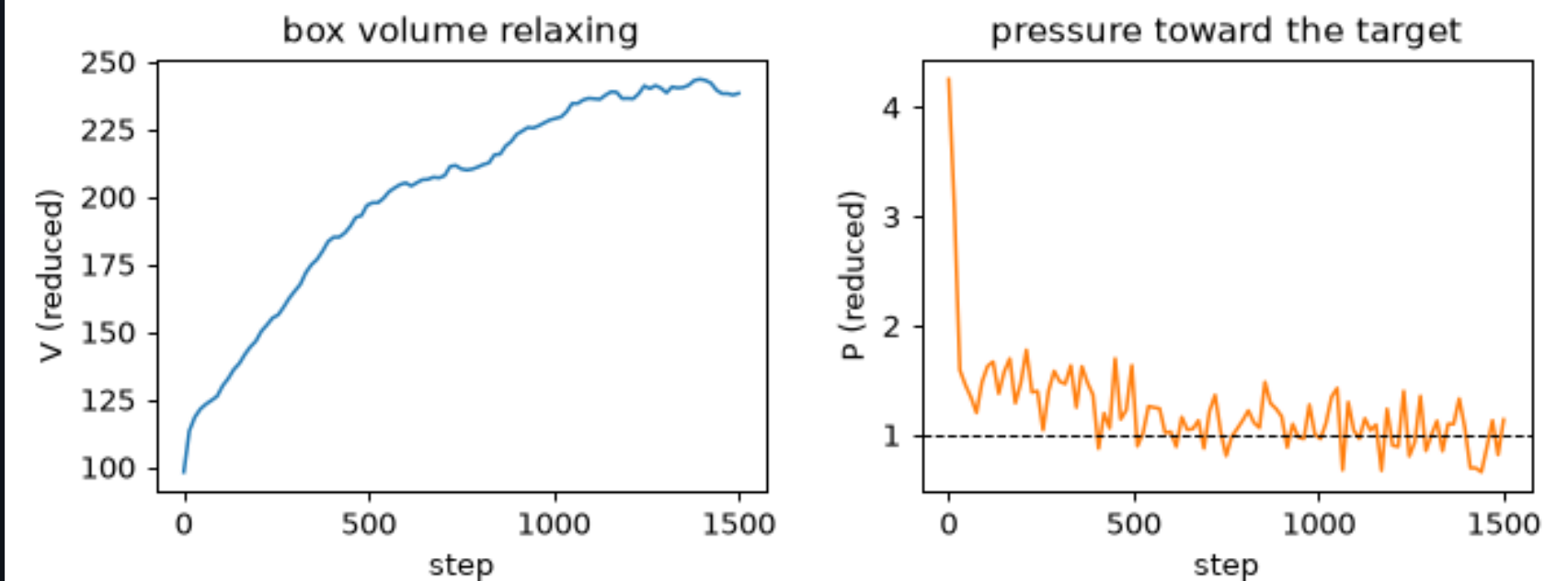
An NPT sketch with its limits stated, LAMMPS's own fix npt, and restarting a run through write_restart / read_restart.

1 Beyond NVT: pressure, and picking a run back up

Beyond NVT: pressure, and picking a run back up

NPT couples a barostat to the thermostat so the box itself relaxes toward a target pressure; restarting means LAMMPS can continue a run exactly, not merely resume from the last log line.

- The `mdlite` NPT here is a stated sketch (Berendsen barostat only) -- good enough to show the mechanism, not a substitute for LAMMPS's `fix npt`.
- `write_restart` / `read_restart` carry the box and every atom's state; a restart script cannot redeclare `units`, `dimension` or `boundary` once the box exists.
- This is the pattern for a run too long to finish in one session: checkpoint, then continue.



The NPT run started denser than its target pressure: box volume expands (left) while the instantaneous pressure relaxes down toward the target of 1.0 (right, dashed) under the Berendsen barostat.

– Figure 1 · notebook §0 · chapter 4, cell 5

Potentials: EAM

An embedded-atom potential read from a real setfl file: copper's lattice constant and cohesive energy against the recorded $3e-5$ -- and why the potential file is never shipped.

1 A real potential, from a real file

A real potential, from a real file

The embedded-atom method adds a density-dependent term to a pairwise potential -- realistic enough for metals, and never something this project ships.

- The setfl file is fetched by the installer from the official source, never tracked in this repository (rule 7).
- From it, LAMMPS and `mdlite` both reproduce copper's lattice constant a_o and cohesive energy E_{coh} , agreeing to 3×10^{-5} .
- This is the forum's other recurring beginner question -- wrong pair_coeff, a potential file that silently does not match the element -- answered by a case built the same checked way as chapter 1.

Structure and Analysis

Radial distribution function, structure factor, mean-squared displacement, diffusion coefficient and velocity autocorrelation, computed from one trajectory.

1 Turning a trajectory into physics

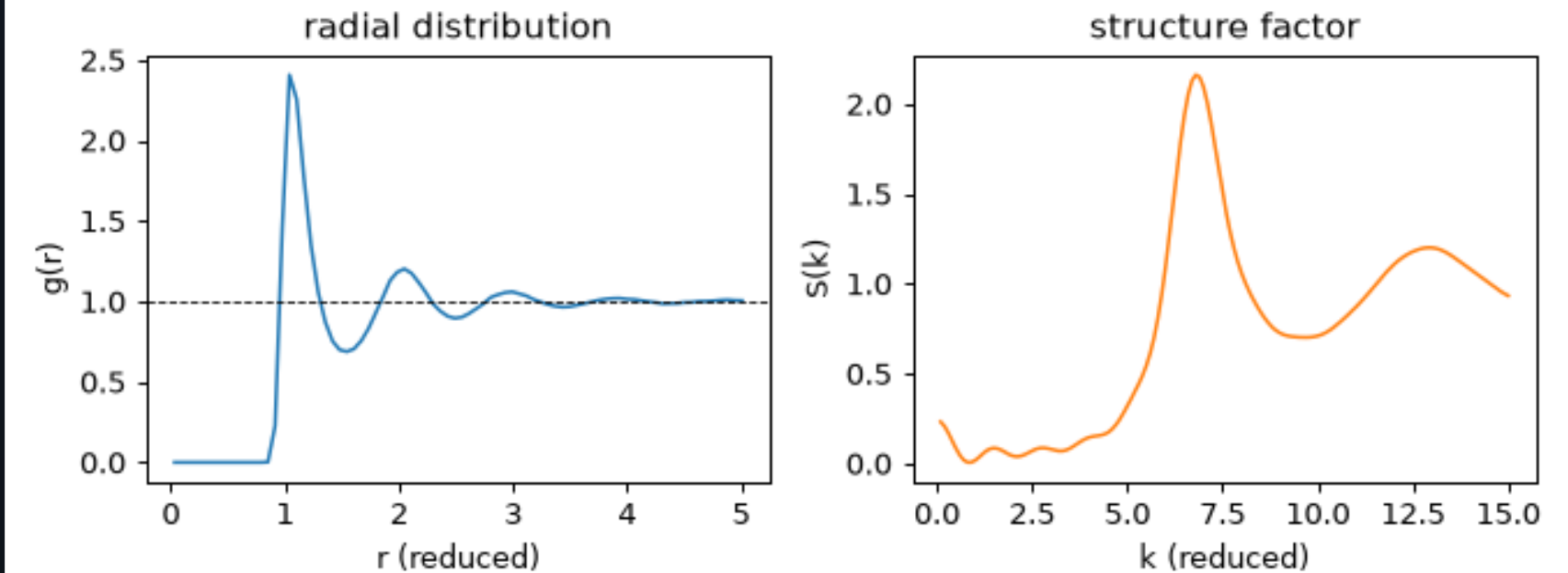
2 From a trajectory to a diffusion coefficient

Turning a trajectory into physics

A dump file is a sequence of raw coordinates;

`lammpskill.post` turns it into the quantities a paper would actually report.

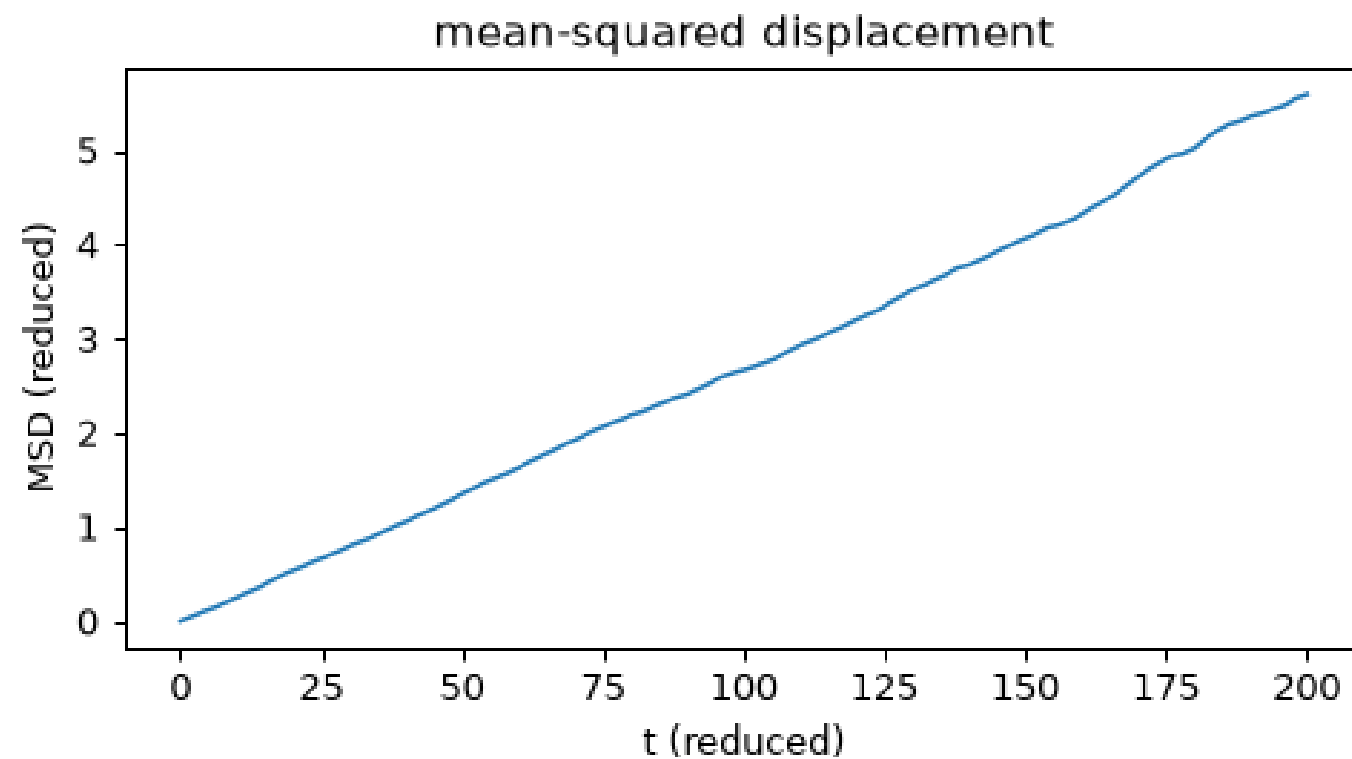
- $g(r)$ and $S(k)$ describe the same structure in real and reciprocal space -- one computed from the other, not measured independently.
- MSD's late-time slope gives a diffusion coefficient by the Einstein relation; block averages give it an honest error bar.
- The velocity autocorrelation function distinguishes a liquid (decays monotonically) from a solid (oscillates, atoms caged) -- itself a check that the case behaved as intended.



The liquid's structure in real and reciprocal space from the same trajectory: the radial distribution function $g(r)$ (left) with its first-neighbour peak and decay to the ideal-gas value of 1, and the structure factor $S(k)$ (right) computed from it. – Figure 1 · notebook §0 · chapter 6, cell 6

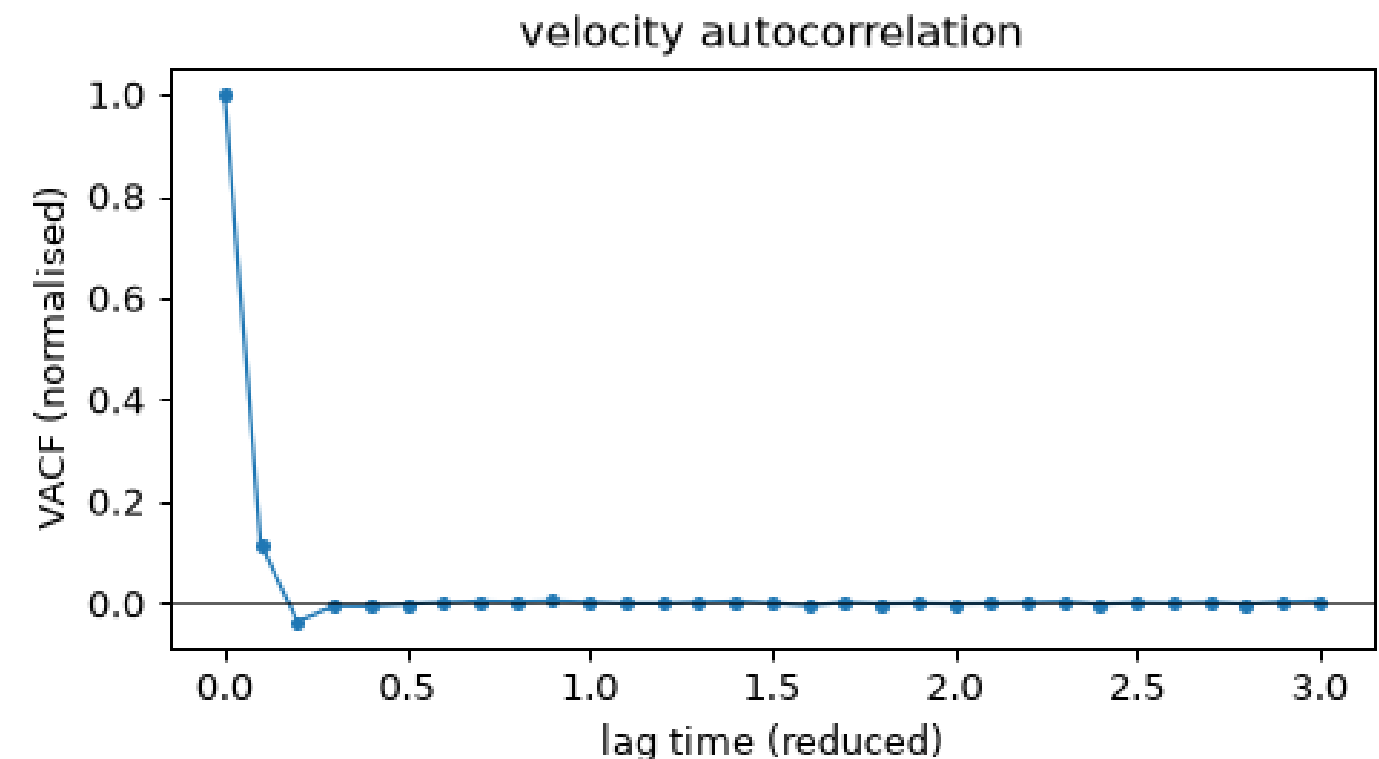
From a trajectory to a diffusion coefficient

The same run, seen two more ways: how far particles have moved (MSD) and how quickly they forget their own velocity (VACF) -- both routes to the same physics.



Mean-squared displacement against time: the slope of its later, genuinely diffusive part (not the early ballistic rise) is the Einstein-relation fit `lammps_kill.post.diffusion_coefficient` reports as `D`.

– Figure 2 · notebook §0 · chapter 6, cell 8



The velocity autocorrelation function, normalised to 1 at zero lag: a liquid decays roughly monotonically toward zero rather than oscillating, which is itself evidence the case really is a liquid.

– Figure 3 · notebook §0 · chapter 6, cell 10

- `lammps_kill.post.diffusion_coefficient` fits the slope over the second half of $\text{MSD}(t)$, where the motion is genuinely diffusive.
- The VACF's decay to zero and the MSD's crossover from ballistic to diffusive are two views of the same relaxation time.

Minimisation

Steepest descent against FIRE on an LJ wall -- and the lesson that gives lost atoms their name: an uncapped step explodes at $f_{\max} 2.6e14$.

1 Why atoms get "lost"

Why atoms get "lost"

The single largest category of beginner questions on the LAMMPS forum is lost atoms and nan energies (docs/08) -- this chapter reproduces the mechanism directly.

- A fixed-size steepest-descent step on an LJ wall explodes: the recorded force reaches $f_{max} = 2.6 \times 10^{14}$ before the run gives up (finding N-7).
- FIRE (adaptive step, velocity-aware) converges the same case without blowing up.
- The fix that matters in practice is not a better algorithm but a **displacement cap** -- bound how far one step is allowed to move an atom, whichever minimiser you use.

Reading What LAMMPS Writes

Data, dump, log and restart formats, and four pitfalls this project hit first: thermo normalisation, dump precision, the restart magic number, multi-format logs.

1 Four formats, four pitfalls

Four formats, four pitfalls

Every format LAMMPS writes has one behaviour that looks like a bug the first time you meet it.

Format	The pitfall this project hit
log (thermo)	<code>thermo_modify norm</code> defaults to per-atom in lj units -- an energy column can look off by the atom count until you know that
dump	float precision is truncated by default; a coordinate that looks identical to the data file may not be, bit for bit
restart	carries an undocumented 16-byte magic number this project had to observe directly rather than find in the manual
log (multi style)	the <code>multi</code> thermo style's block layout parses differently from the default one-line-per-step style

Choosing a Build

What a 314-case sweep of the examples says: package list beats package count, and how to read `Installation.packages` before choosing styles.

1 The build you have decides what you can run

The build you have decides what you can run

A 314-case sweep of LAMMPS's own example scripts, run on this project's two installed builds, is the evidence behind this lecture.

- Package `list` beats package `count`: an 11-package build ran cases a 52-package build could not, because the specific packages differed.
- `lammpsinstall.Installation.packages` is read from `lmp -h`, not assumed from the build recipe -- the same question ("is the package actually in my build?") the MOLFIE forum thread asked once, answered by measurement.
- The Windows installer is the single most-viewed installation thread of the year on the LAMMPS forum -- and the one route this project cannot yet speak to from its own measurements (open question Q-1).

Scaling and Limits

MPI and OpenMP speed-up on eight cores, where it stops, and what this machine honestly cannot do -- no GPU, no -partition methods.

1 Where speed-up stops, and what this machine cannot do

Where speed-up stops, and what this machine cannot do

Measured on this project's 8-core, no-GPU machine: 2 MPI ranks gave 1.73x, 2 OpenMP threads gave 2.09x -- both honestly short of ideal.

- Parallel efficiency drops as communication overhead grows relative to the per-rank workload -- expected, and worth showing rather than assuming.
- This machine has no GPU and no `-partition` support, so KOKKOS/GPU acceleration and multi-replica methods (NEB, TAD) are out of scope here by design, not by oversight.
- The honest version of a scaling claim states the hardware it was measured on -- these numbers are this machine's, not a general LAMMPS property.